

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)	(i)		Enzyme-substrate complex		1		1		
		(ii)		Glycerol		1		1		
	(b)	(i)		0.11		1		1	1	1
		(ii)		Both scales correct + Y axes label = rate of reaction (1) All plots correct = 2 marks $\pm$ <1 small square 4 plots correct = 1 mark 0/1/2/3 plots correct = 0 marks Line joining plots (1) $\pm$ <1 small square	1	2 1		4	4	
		(iii)		As {temp/ it} increases rate of reaction increases to <u>40(°C)</u> / <u>0.21</u> and then decreases/ the rate of reaction peaks at <u>40(°C)</u> / <u>0.21</u> (1)		1		1		1
		(iv)		More kinetic energy/ increases movement of enzymes (1) More (successful) {collisions/ enzyme substrate complexes} (1)		2		2		
	(c)			To come to the {working/reaction} temperature/ to acclimatise/ equilibrate/ come to the same temperature as the water bath Ignore to get used to the temperature/ to warm up			1	1		
	(d)			Low in fat / does not contain (enough) fat/ as the fat in milk is needed for the experiment/ Only 1% fat		1		1		
				Question 6 total	1	10	1	12	5	3